

## Motor Subsystem Theory and Implementation

### Theory

The motor of the buggy can be modelled like in figure 1 as an inductor and resistor in series with back emf. Equation {1} can be derived and therefore see how each element relates to the input voltage of the motor.

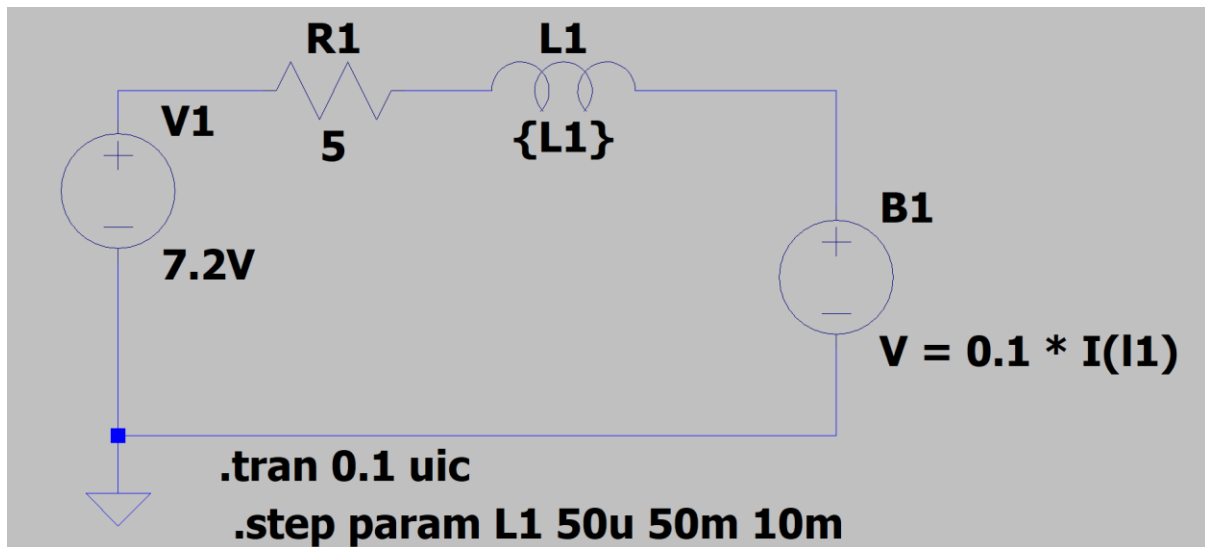


Figure 1: Basic motor model in LTspice

Here the motor torque constant from equation {2} is set arbitrarily to 0.1 Nm/A. When simulating like in figure 2, with different inductance values, the current rise is much slower for larger values of L. Equation {2} shows that a greater current will allow a greater torque, so it is desirable to have a faster current rise time for the instantaneous current. This is because the induced current acts as a back emf opposing motion of the motor.

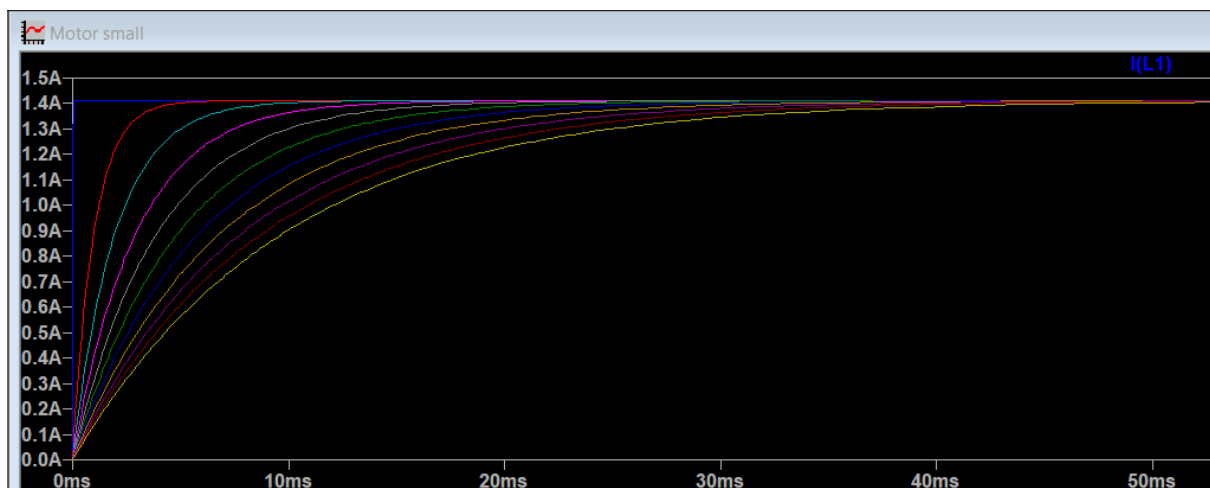


Figure 2: Simulation of inductance effects

The motor is powered by PWM (Pulse Width Modulation) signals. This is where the voltage is rapidly turned on and off (High and Low) to create pulses/square waves as seen in figure 1.

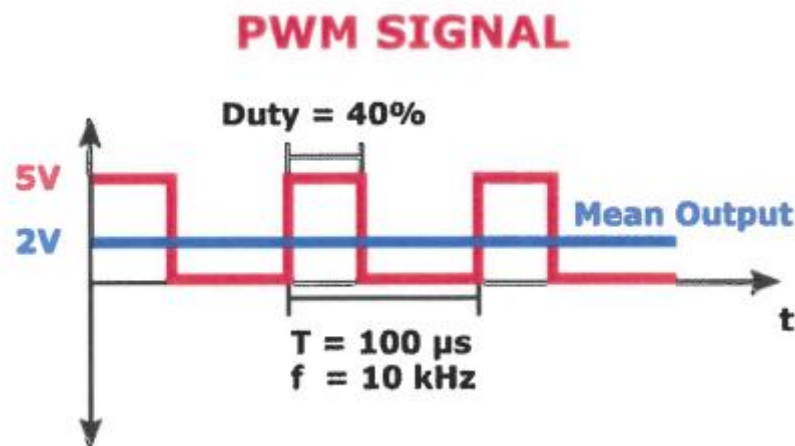


Figure 3: PWM average voltage

The duty cycle is the pulse width (in time) / total period of the signal as a percentage, which basically is how long the signal is high compared to the overall signal.

The frequency is how fast the signal switches between high and low. This can affect shoot through issues which will be revisited later.

Average voltage = (Duty Cycle/100) \* Maximum Voltage. Therefore, we can change the duty cycle to attain desired voltages needed. When we have a higher duty cycle, the average voltage increases therefore the speed of the motor also increases. But why do we need to use PWM instead of a constant voltage? By sending pulses, the heat loss is far less than if linear voltage was used i.e. more energy efficient [1]. Since Torque is proportional to current {1}, when PWM is used the speed of the motor can be controlled without lowering the torque because the max current is preserved (Average current may be dropped but torque only relies on the instantaneous current [2]. With linear voltage, this is not possible.

### H bridge theory

The H bridge allows Bi-directional control of the motor and uses PWM signals to control the speed of the motor. This works by having two N-MOSFET's and two P-MOSFET's that act as switches arranged like in figure 4.

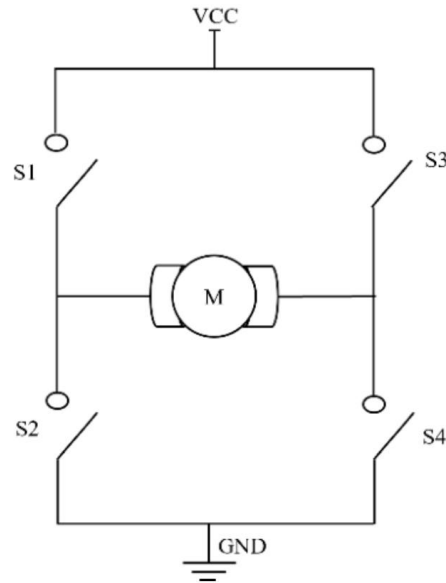


Figure 4: MOSFET arranged in a H-bridge shown as switches.

When sending a PWM signal, it oscillates between High and LOW signals which determines which MOSFET on the half bridge (one side of the H bridge) is turned on and off. With the battery voltage being 7.2V and the max voltage of the PWM being 5V, these conditions will determine how the MOSFET behaves.

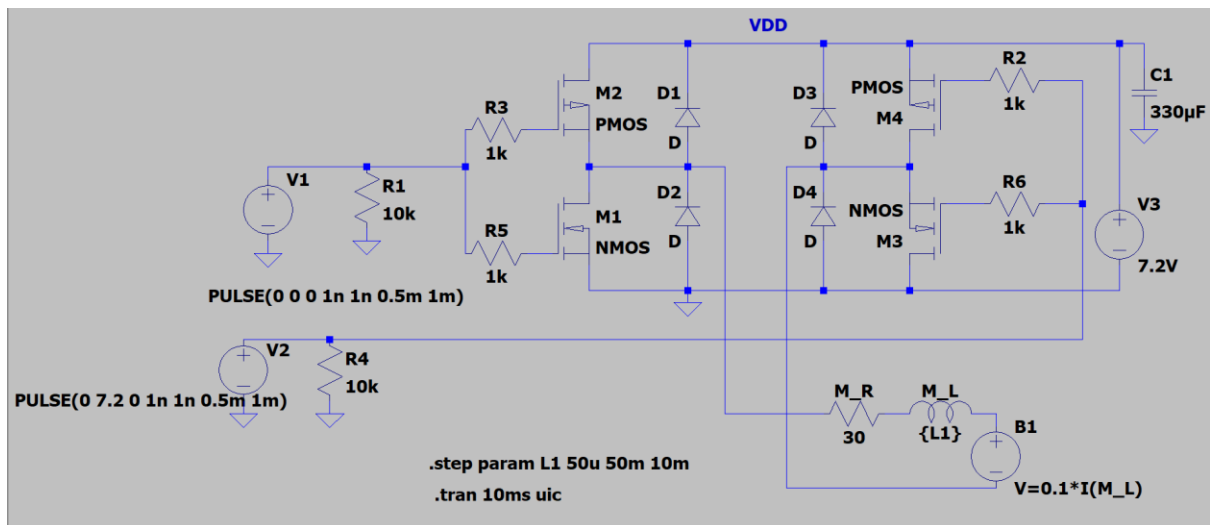


Figure 5: Design of H bridge using LT spice

Figure 5 shows the design of the H bridge ready for simulation. V1 is set to a low signal (0) whilst V2 is set to a pulse signal of 7.2V (Since in practice using a gate driver it will be amplified), to mimic the PWM signals.

M1 and M3 are connected to ground therefore when the PWM V2 signal is low, the NMOS' are switched off and when the signal is high, the NMOS' are switched on. The PMOS and NMOS switch complementary to each-other.



Figure 6: Circuit when V1 low and V2 high

Figure 6 shows how the complementary switching of the MOSFETs enable current flow and get the motor spinning. In the figure 5 example, only the right half bridge is switching whilst the left-hand side is staying constant. This is a unipolar drive, where M2 is constantly on and M1 is constantly off [3]. Only the right half bridge is switching with the signal V2, so therefore when M3 is on then current flows hence the motor rotates clockwise. When M4 is on then there is a braking operation. Since there is no voltage applied to the motor as there is no connection to ground, the motor's back EMF opposing motion helps slow itself down. This induced EMF needs a safe passage to ground without damaging the circuit which will be discussed later. The motor cannot be reversed in a unipolar drive, but direction of the motor depends on if we manually set V1 or V2 to low.

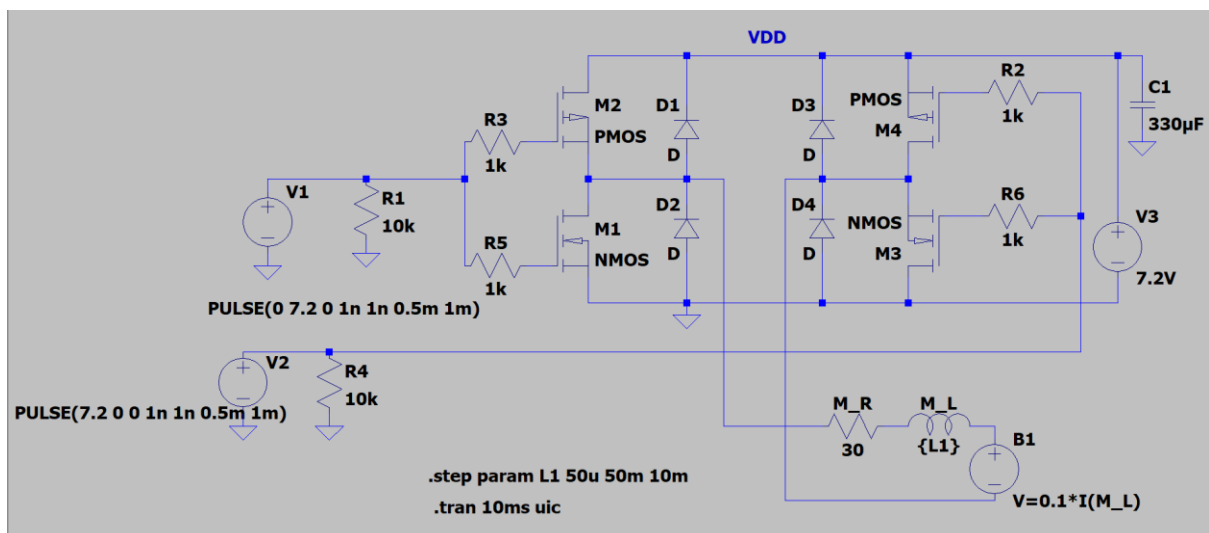


Figure 7: Bipolar drive in LT spice

Figure 7 shows the bipolar drive that can reverse the motor. In this example, V1 and V2 are sending pulses with a phase difference of 180°. This means that each half bridge is

also switching complementary to each-other. The signals should make the motor rotate clockwise and anticlockwise very quickly however due to inertia the motor can keep spinning in one direction. From the motor's perspective, it is switching between 7.2V to -7.2V therefore the change in VDD is 14.4V instead of 7.2V in a unipolar drive. Equation {1} shows that larger V is proportional to greater change in current per time therefore using equation {2} means that the torque is doubled when using bipolar drives. In theory, the average voltage is 0 when there is a complementary V1 and V2 signal so to bias a direction the PWM duty cycle can be initially unbalanced where the high of V1 and the low of V2 have a different duty cycle to the low of V1 and high of V2 as seen in figure 8.

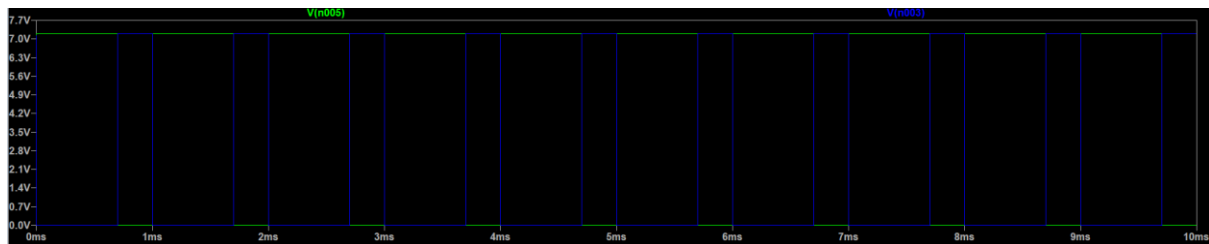


Figure 8: V1 (green) and V2 (blue) with varying duty cycles to bias motor initially.

Using Bipolar drives can come with more challenges such as shoot through. This is when both MOSFETs in the half bridge are switched on creating a short in the circuit and can damage the MOSFETs. This can occur if the half bridges overlap in their switching, so no current flows through the motor and a short is created like in figure 9.

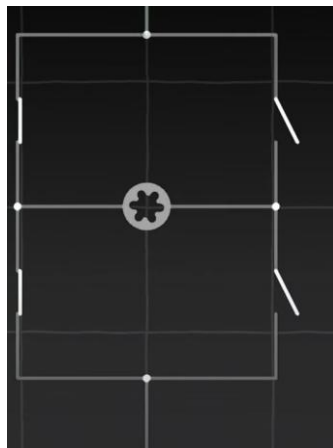


Figure 9: Short circuit creating spikes in current

D1, D2, D3 and D4 (from figure 5) are flyback diodes designed to provide an alternative pathway to induced current by the motor. When the motor is turned off after being turned on, it will keep spinning which induces a current [4]. The diodes provide a path for this current to safely reach ground without damaging the components. This can be seen in figures 10 and 11.

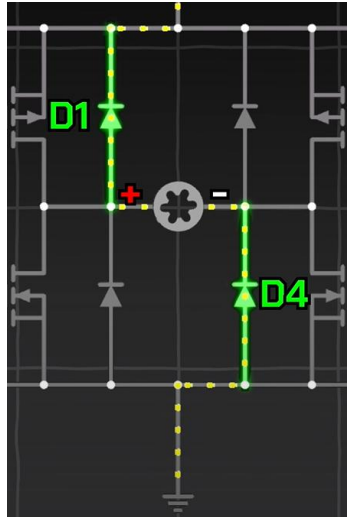


Figure 10: Safe induced current pathway

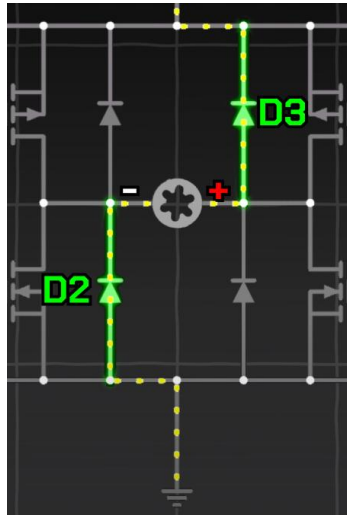


Figure 11: Safe current pathway 2

To eliminate shoot-through, delays between switching must be added between turning off the PMOS and turning on its complement NMOS and vice versa. R3, R5, R2 and R6 are current limiters for the MOSFETs, which protect them in case of surges of current. They also affect the switching time of the MOSFET, with higher resistances meaning slower switching [5].

R1 and R4 are pull down resistors. This is so that if there is any Electromagnetic interference or other current spikes then there is no accidental turn on of the MOSFETs that could lead to shoot through. It makes sure that it will drop the current to as close as zero as possible.

C1 and C2 are used for filtering noise. C1 is decoupling and smoothing large changes in the supply voltage due to PWM switching and motor inductance pulling lots of current.

C2 was a 0.1uF capacitor connected to VDD and ground and is used to absorb dips and oscillations in the battery voltage when the MOSFETs are switching at high speeds and

are charging/discharging. It is used to keep the supply voltage stable. They have a large difference in values as C1 is designed for big swings in voltage whilst C2 is designed for fast, small oscillations in the supply [6].

The gate driver was not included in simulation however this is used to amplify the maximum 5V from the microcontroller to 7.2V. It also helps with switching speeds and inverts the signals. It will take two signals to send to the half bridges.

### Testing

A Unipolar drive was made on breadboard by grounding the left PMOS forcing it on. Without C2, the signal motor was receiving looked like figure 12.



Figure 12: Input of motor without C2

Figure 13: shows the PWM the motor receives without C1.



Figure 13: Input of motor without C1

Figure 14 shows the signal when we include both capacitors and can see that the noise has been decoupled successfully.

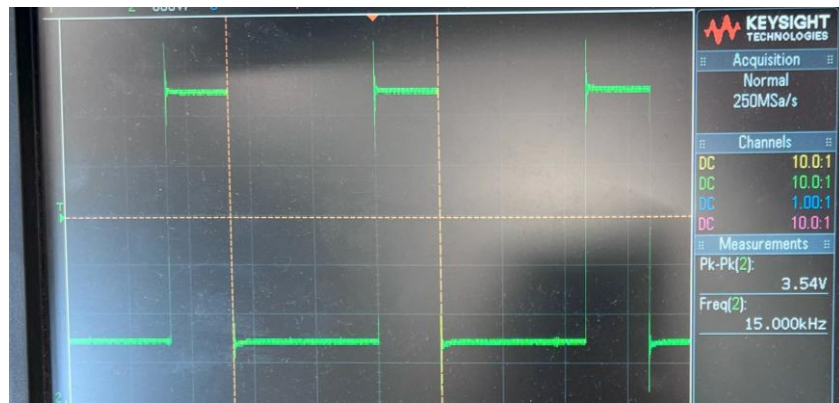


Figure 14: Input of motor with C1 and C2

Figure 15 shows the breadboard designed and built for testing.

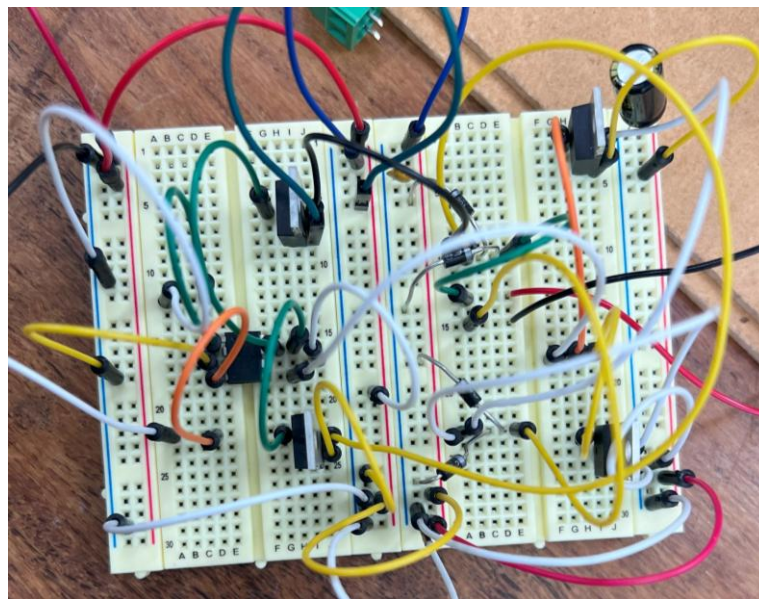


Figure 15: Breadboard design of H bridge

After numerous tests, it was found that the current limiters did not have much effect on the switching delays and speeds as the gate driver already handles that so the stripboard in figure 16 omits these.

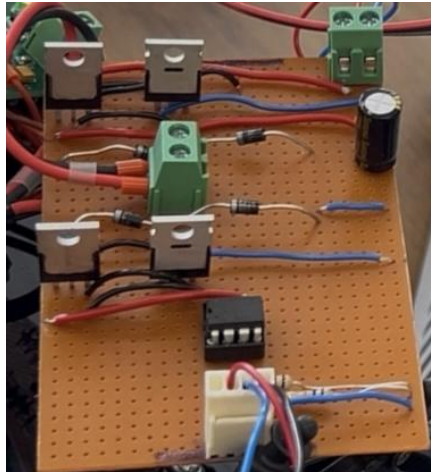


Figure 16: Stripboard design

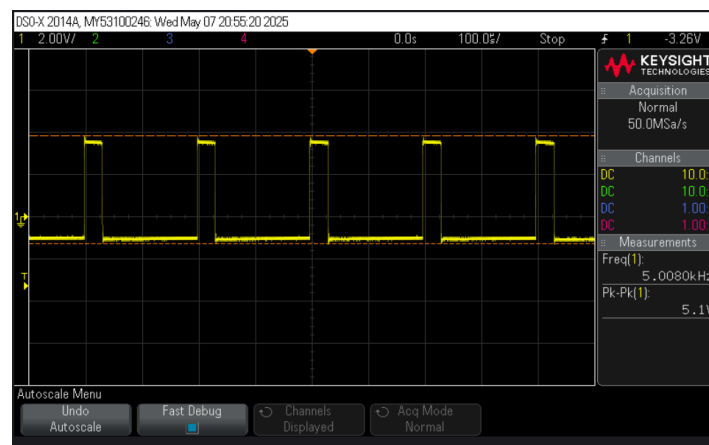


Figure 17: Pin 4 gate driver, RA4 input [7]

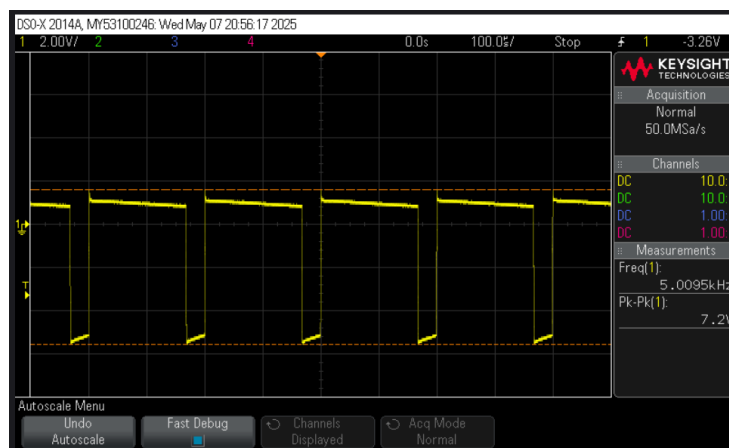


Figure 18: Pin 5 gate driver, RA4 output [7]

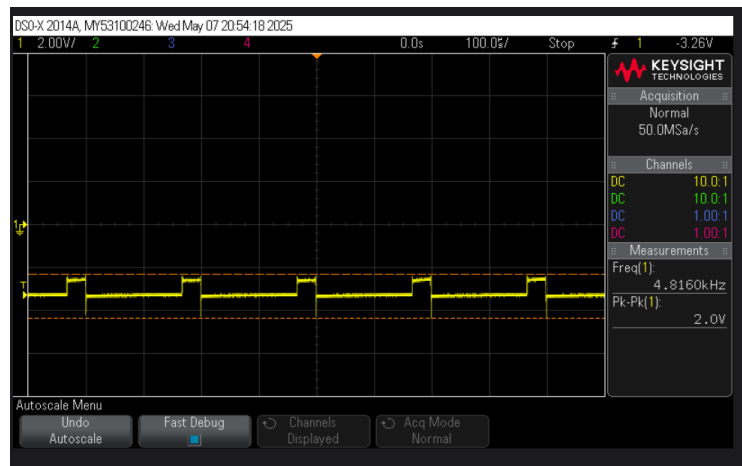


Figure 19: Pin 2 gate driver, RA5 input [7]

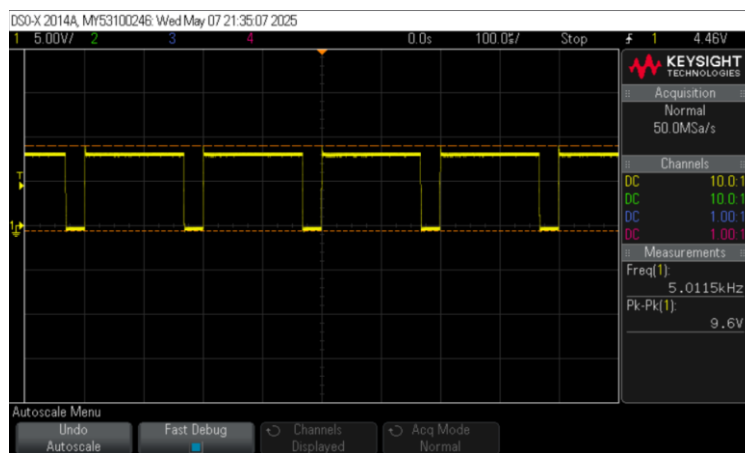


Figure 20: Pin 7 gate driver, RA5 output

Figures 17-20 show how the gate driver amplifies the signal from the output of the PIC.

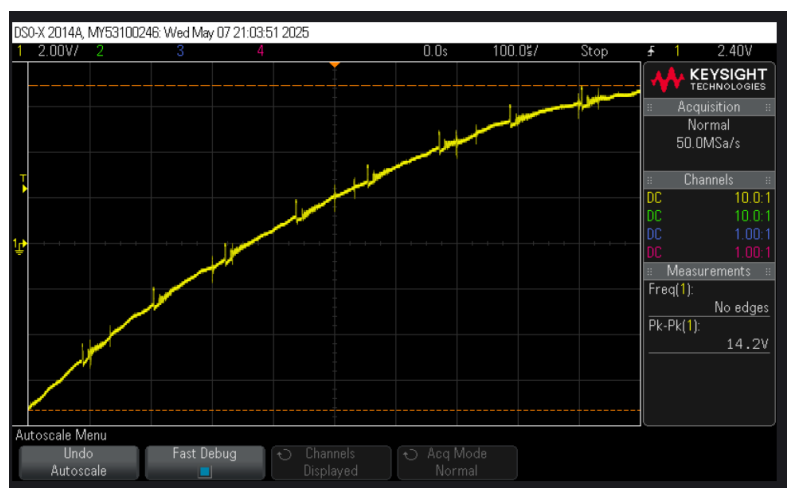


Figure 21: PMOS gate voltage

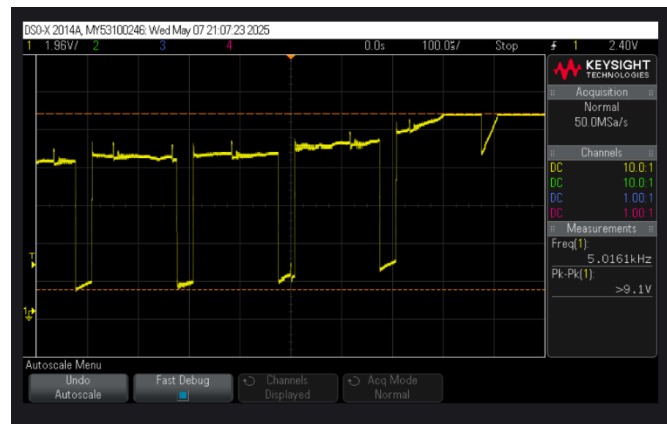


Figure 22: NMOS gate voltage

Figure 21 and 22 shows the gate voltage and the voltages when switching on and off as discussed in the theory.

At a 30% duty cycle, the motor was run at 1kHz to 750 kHz and found the ideal PWM signal should be 100kHz for fastest results. The H-bridge has a minimum operating voltage of 4.9V for the local supply voltage.

$$\{1\} V = e + (i * Ra) + (La * \frac{dia}{dt}),$$

Voltage = Back emf voltage + (current \* motor resistance) + (motor inductance \* change in current/ time)

$$\{2\} T = k_t * I,$$

Torque = motor torque constant \* current.

$$\{3\} e = Ke * \omega r = Ke \frac{d\phi r}{dt}$$

Back emf = Back emf constant \* rotor velocity = Back emf constant \* (change in rotor position / time)

$$\{4\} kt = ke$$

Motor torque constant = Back emf constant

[1]

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